EXP NO :-9
PROGRAM:-

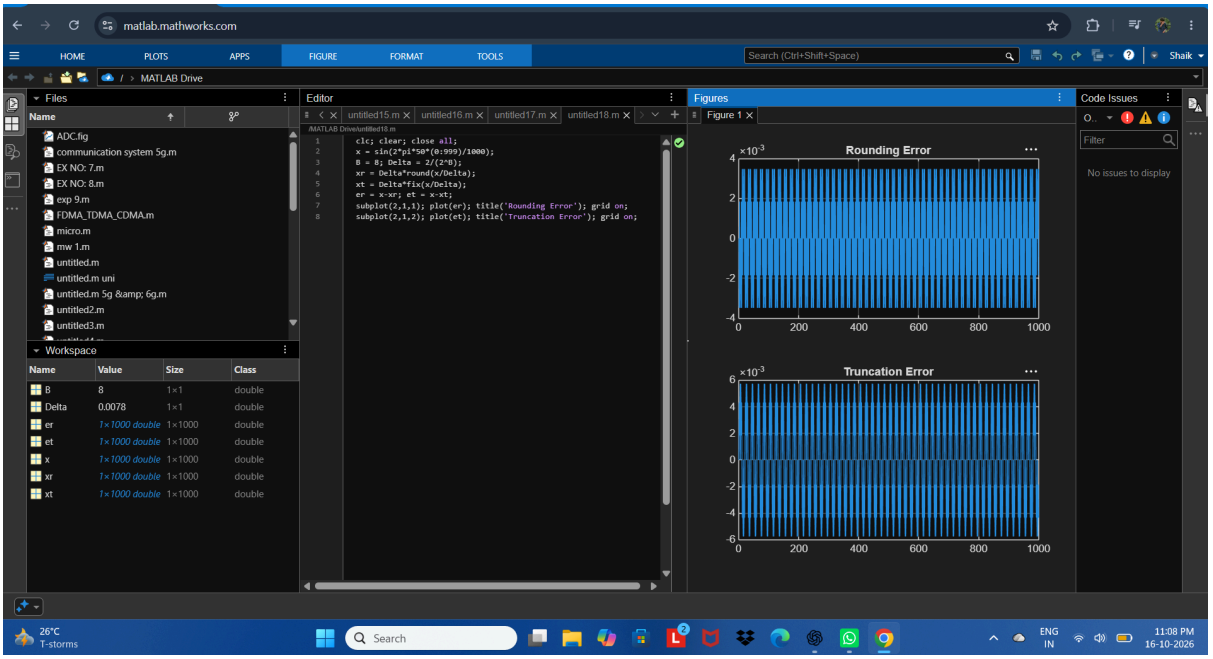
```
clc; clear; close all;
n = 0:999; x = sin(2*pi*50*n/1000);
for B = [4 8 12]
    Delta = 2/(2^B);
    xq = Delta*round(x/Delta);
    plot(n(1:100),xq(1:100)); hold on;
end
legend('4-bit','8-bit','12-bit');
xlabel('Sample'); ylabel('Amplitude'); grid on;
```

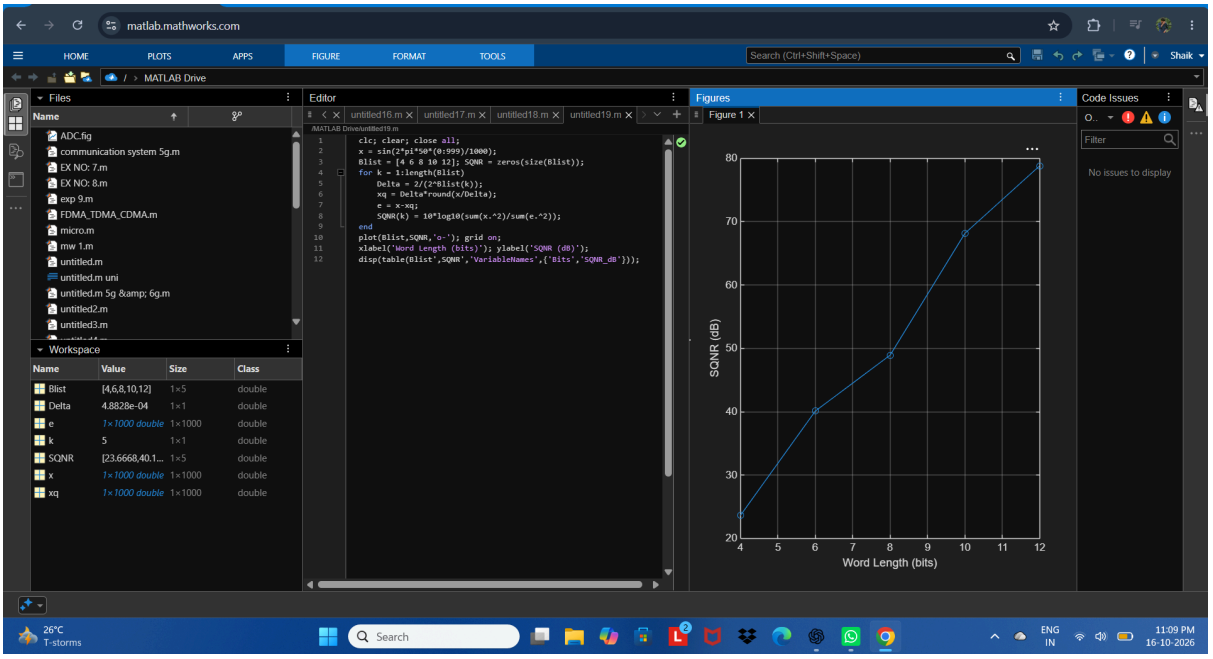
```
clc; clear; close all;
x = sin(2*pi*50*(0:999)/1000);
B = 8; Delta = 2/(2^B);
xr = Delta*round(x/Delta);
xt = Delta*fix(x/Delta);
er = x-xr; et = x-xt;
subplot(2,1,1); plot(er); title('Rounding Error'); grid on;
subplot(2,1,2); plot(et); title('Truncation Error'); grid on;
```

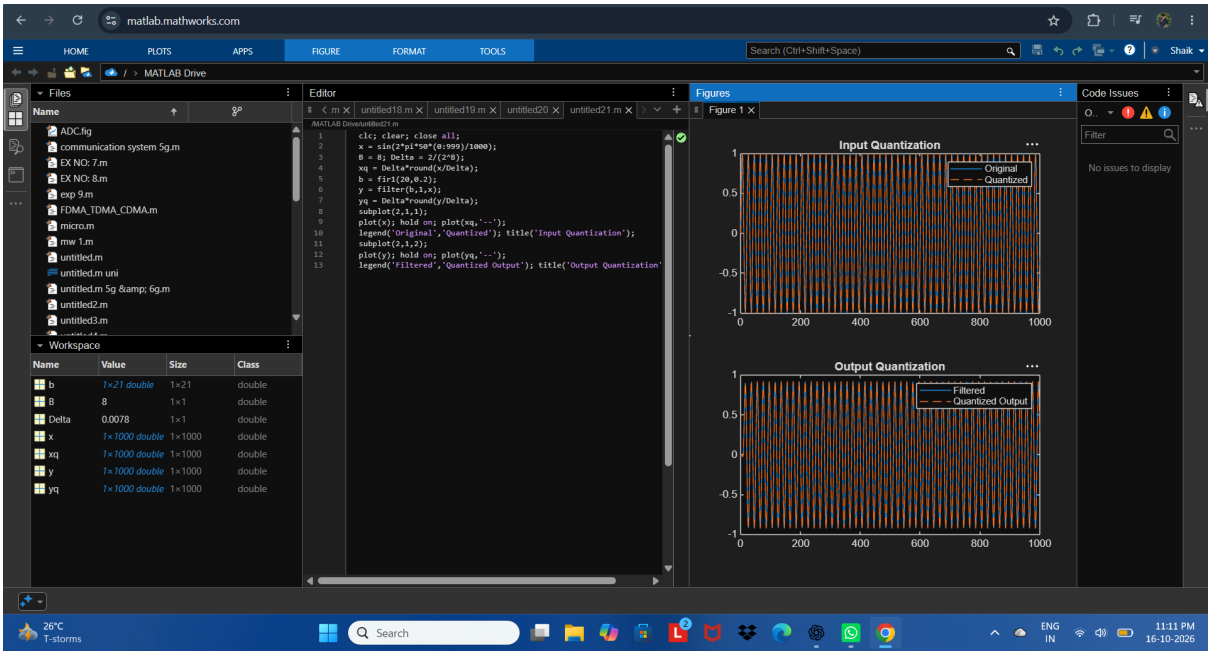
```
clc; clear; close all;
x = sin(2*pi*50*(0:999)/1000);
Blist = [4 6 8 10 12]; SQNR = zeros(size(Blist));
for k = 1:length(Blist)
    Delta = 2/(2^Blist(k));
    xq = Delta*round(x/Delta);
    e = x-xq;
```

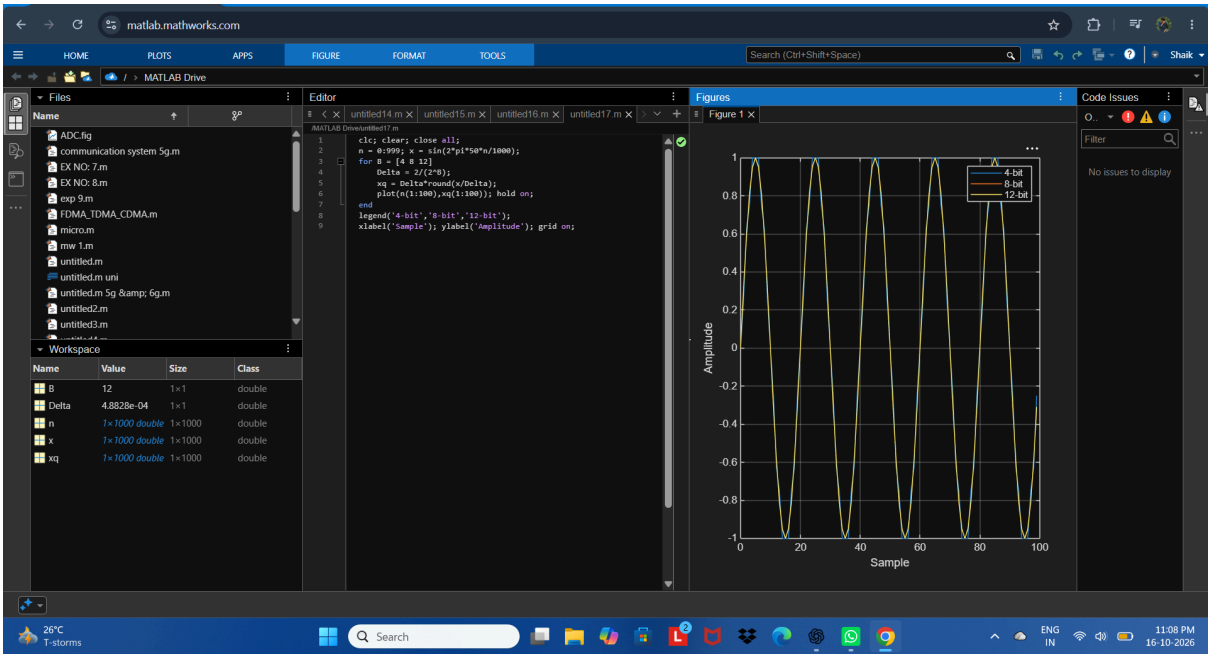
```
    SQNR(k) = 10*log10(sum(x.^2)/sum(e.^2));
end
plot(Blist,SQNR,'o-'); grid on;
xlabel('Word Length (bits)'); ylabel('SQNR (dB)');
disp(table(Blist',SQNR','VariableNames',{'Bits','SQNR_dB'}))
;

clc; clear; close all;
x = sin(2*pi*50*(0:999)/1000);
B = 8; Delta = 2/(2^B);
xq = Delta*round(x/Delta);
b = fir1(20,0.2);
y = filter(b,1,x);
yq = Delta*round(y/Delta);
subplot(2,1,1);
plot(x); hold on; plot(xq,'--');
legend('Original','Quantized'); title('Input Quantization');
subplot(2,1,2);
plot(y); hold on; plot(yq,'--');
legend('Filtered','Quantized Output'); title('Output
Quantization');
```


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Name
ADC.fig
communication system 5g.m
EX NO: 7.m
EX NO: 8.m
exp 9.m
FDMA_TDMA_CDMA.m
micro.m
mw 1.m
untitled.m
untitled.m uni
untitled.m 5g & 6g.m
untitled2.m
untitled3.m
Workspace
Name Value Size Class
B 8 1×1 double
Delta 0.0078 1×1 double
er 1×1000 double 1×1000 double
et 1×1000 double 1×1000 double
x 1×1000 double 1×1000 double
xr 1×1000 double 1×1000 double
xt 1×1000 double 1×1000 double
Editor
untitled15.m
untitled16.m
untitled17.m
untitled18.m
clc; clear; close all;
x = sin(2*pi*50*(0:999)/1000);
B = 8; Delta = 2/(2^B);
xr = Delta*round(x/Delta);
xt = Delta*fix(x/Delta);
er = x-xr; et = x-xt;
subplot(2,1,1); plot(er); title('Rounding Error'); grid on;
subplot(2,1,2); plot(et); title('Truncation Error'); grid on;
Figures
Figure 1
Rounding Error
Truncation Error
Code Issues
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Name
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EX NO: 8.m
exp 9.m
FDMA_TDMA_CDMA.m
micro.m
mw 1.m
untitled.m
untitled.m uni
untitled.m 5g & 6g.m
untitled2.m
untitled3.m
Workspace
Name Value Size Class
Blist [4,6,8,10,12] 1×5 double
Delta 4.8828e-04 1×1 double
e 1×1000 double 1×1000 double
k 5 1×1 double
SQNR [23.6668,40.1... 1×5 double
x 1×1000 double 1×1000 double
xq 1×1000 double 1×1000 double
Editor
untitled16.m
untitled17.m
untitled18.m
untitled19.m
clc; clear; close all;
x = sin(2*pi*50*(0:999)/1000);
Blist = [4 6 8 10 12]; SQNR = zeros(size(Blist));
for k = 1:length(Blist)
Delta = 2/(2^Blist(k));
xq = Delta*round(x/Delta);
e = x-xq;
SQNR(k) = 10*log10(sum(x.^2)/sum(e.^2));
end
plot(Blist,SQNR,'o-'); grid on;
xlabel('Word Length (bits)'); ylabel('SQNR (dB)');
disp(table(Blist',SQNR','VariableNames',{'Bits','SQNR_dB'}));
Figures
Figure 1
SQNR (dB)
Word Length (bits)
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Workspace
clc; clear; close all;
x = sin(2*pi*50*(0:999)/1000);
B = 8; Delta = 2/(2^B);
xq = Delta*round(x/Delta);
b = fir1(20,0.2);
y = filter(b,1,x);
yq = Delta*round(y/Delta);
subplot(2,1,1);
plot(x); hold on; plot(xq,'--');
legend('Original','Quantized'); title('Input Quantization');
subplot(2,1,2);
plot(y); hold on; plot(yq,'--');
legend('Filtered','Quantized Output'); title('Output Quantization')
Input Quantization
Original
Quantized
Output Quantization
Filtered
Quantized Output
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Delta = 2/(2^B);
xq = Delta*round(x/Delta);
plot(n(1:100),xq(1:100)); hold on;
end
legend('4-bit','8-bit','12-bit');
xlabel('Sample'); ylabel('Amplitude'); grid on;
4-bit
8-bit
12-bit
Amplitude
Sample
No issues to display
26°C
T-storms
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